Evaluation of Weather Forecast with DL

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***Abstract*—Widespread availability of weather observation data sets in the last decade and the research and techniques available in deep learning sector have motivated many researchers and to explore hidden hierarchical patters in vast quantities from the available data sets. The goal of meteorological scientists has always been to anticipate the weather accurately and in a timely manner. Traditional theory-driven numerical weather prediction systems face many challenges including a lack of understanding of physical causes, difficulties in extracting usable information from a flood of observation data, and the need for significant computational resources. The successful implementation of data-driven deep learning methods in a variety of domains, including computer vision, speech recognition, and time series prediction, has demonstrated that deep learning methods can efficiently extract temporal and spatial features from spatio-temporal data. [1] According to research, there is some evidence present which suggests that better weather forecasts can be produced by the introduction of big data mining and neural networks into the weather prediction workflow. [2]**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

Due to its impact on worldwide human life, weather forecasting has attracted the interest of numerous scholars from diverse research areas. The histories of numerical weather prediction (NWP) and machine learning (ML) or artificial intelligence (AI) (for the purposes of this research, the names are interchangeable) are vastly different. Lewis Fry Richardson explored manual NWP in Britain in 1922, and the first computer-aided weather forecasts were published in 1950. [3]

Operational weather forecasts in Sweden, the United States, and Japan followed soon after. While the evolution of NWP has been steady [4], the history of ML [5] has been more disruptive: McCulloch & Pitts proposed the first neural network (NN) in 1943. [6] The subject developed until the early 1960s, when existing algorithms were discovered to be inefficient and unstable. Backpropagation's development in 1970 [7], [8] ushered in a second wave of machine learning applications, as it became possible to build bigger NNs and train them to recognize nonlinear correlations in data. [9] Even though ML algorithms were still being developed, interest in them declined because they rarely showed significant performance gains and computing resources were insufficient to address larger issues. Large amounts of labelled data, which are required for most data-driven ML techniques, were also difficult to come by.

Weather forecasting is a vast field of research with wide potential applications which range from flight navigation, tropical cyclones, agriculture and travelling expeditions. Constraints of weather forecasting include learning of massive volume datasets and then developing an executable weather prediction model which revolves around learning structural patterns in that huge data set.

In recent times, alot of research has happened in forecasting weather using statistical modeling techniques including machine learning which have provided successful results. [10]–[12]. Significant developments around 2010 kicked off the third wave of artificial intelligence, which is still going strong today: computing capabilities were vastly expanded thanks to massive parallel processing in graphical processing units (GPUs), convolutional neural networks (CNN) allowed for much more efficient analysis of massive (image) datasets, and large benchmark datasets were made available on the internet.

The studies in the fields of deep belief nets [13], deep networks [14], energy-based models [15] have acted as the base and foundation of deep learning approach as the deep architecture of generative models in the recent decade.The term ”deep” in deep learning refers to a neural network (NN) that has more layers than ”shallow” neural networks utilized in traditional machine learning models. After the successful execution of the layer-wise unsupervised pre-training method, which is used to efficiently handle the training issues, this multi-layer NN has sparked a lot of research interest. When compared to shallow models, the ”deep” architecture is particularly important because NN with deep architecture can provide higher learning ability. [16]

Deep learning applications in various sectors have successfully demonstrated by many researchers, have motivated its utilization in weather forecasting and its modeling. A recent study [17], suggests to represent weather using hierarchical feature that is obtained from a large number of weather data using Deep Neural Network(DNN).I take this opportunity to expand on the study of [18] and investigate the relevance of such methods to the NWP workflow as newer DL methods begin to supply such concepts.

While much work is being done to replace particular sections of the NWP cycle with DL methods. In this paper I would try to investigate whether it is possible to replace all parts of the NWP workflow with one deep learning NN, which would take observations as input and directly produce end-user forecast solutions from the data as illustrated in Fig. 1.

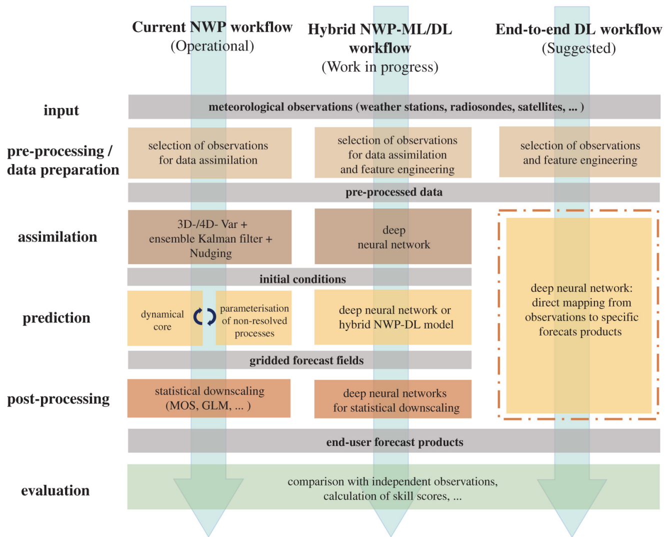

Fig. 1. Idealized workflows of current numerical weather prediction (left), next-generation weather prediction with individual components substituted or augmented by ML and DL techniques (centre), and a purely data-driven DL weather forecasting system (right). [2]

Personal Motivation

The goal of my research is to look at the possibilities of deep learning techniques for weather forecasting using rich hierarchical weather representations learnt from large amounts of weather data. In this paper I shall be using different techniques through which I shall try to evaluate Weather Forecasting using Deep Learning techniques. The discussion will include an analysis of present Machine learning concepts and their applicability to the weather data with its statistical characteristics. Deep Learning based weather prediction is expected to be a strong supplement to the conventional method. In this paper I shall review the state of the art research on deep learning based weather forecasting, from the perspectives of Neural Network architecture design, geographical and temporal scales, data sets, and benchmarks. After that, I shall compare the benefits and drawbacks of Weather forecasting with Deep Learning to traditional Numerical Weather prediction and describe the potential future research subjects for DLWP.

The data set that I am using for the forecast has been collected and manipulated from different resources mainly from Kaggle, FiveThirtyEight, Socrata and Github repositories. The language in which the code is written is Python. For compilation of data I am using Anaconda IDE and Kaggle notebook.

A. Research Aims and Objectives

Modern day weather forecasting greatly relies on the massive simulation systems which work around the clock globally. There are numerous steps involved to attain the complete process chain and all of these steps are interacting closely with each other as it can be seen in the left column of Figure 1.

There are millions of measurements that are obtained round the clock but still they are not sufficient to describe the complete state of the atmosphere and other worldly system components with which the atmosphere around us exchanges its forms of energy. At this point, Data Assimilation comes into play which has just one central task of filling the gap between the random, incomplete and scattered observations which are required by NWP models.

The capacity of NWP models to forecast future atmospheric conditions has steadily increased over the last several decades. Modern global NWP models are capable of forecasting not only the synoptic-scale weather pattern for many days, but also end-user important meteorological parameters such as 2 m temperature and regional-scale precipitation events with surprising precision.

Extensive increase in computational power, availability of larger datasets and the rapid development on new Neural Networks architectures all of this has contributed to the ongoing success of Deep Learning. One popular concept of Neural Networks is Convolutional neural network(CNN) which are a popular and commonly used concept in which a stack of small-sized filters with few trainable parameters is applied to pictures or other gridded input to extract coarser scale characteristics. CNNs have been utilized in weather and climate applications where the neural network (NN) [16] has been trained to detect spatial characteristics, such as in the analysis of satellite data or the output of weather models.

In this paper I will explain all the methodologies that are involved in forecasting weather using deep learning. I will also try to test the data set and with this will in turn prove why with the introduction of deep learning weather forecasting has improved. I will also be highlighting some of the challenges faced by deep learning and at the end I will conclude the paper.

II. FOUNDATION

A. Weather Forecasting Problem

Many significant efforts have been reported in the last decade to solve weather forecasting problem using statistical modeling including machine learning. These reports have resulted in quite efficient results. [19] The input data for forecasting weather is high-dimensional weather time series data which is collected from various weather stations from different regions across the world. Different studies such as study of Recurrent Neural Network(RNN) models to forecast regional annual runoff [20], study by Chen and Hwang [10] which proposes a fuzzy time series model for temperature prediction based on the historical data that are presented by linguistic values. Another research [11] found that an ensemble of artificial neural networks (ANNs) can learn weather patterns from weather parameters. Neural Networks(NN) are used for rainfall forecasting using statistical downscaling. One more research [12] makes use of LIDAR data and proposes a Chaotic Oscillatory-based Neural Network for short term wind forecasting. Integrated ANN and fuzzy logic wavelet model[IO] is deployed to forecast long term rainfall. The analysis with Precipitation index drought forecasting uses

NN, wavelet NN, and Support Vector Regression (SVR). [21] Rainfall forecasting model based on an esemble of artificial NN. [22]

It is very evident that there are multiple studies and methods that are being executed simultaneously for gathering different data and each one outputs information for individual branch of weather. Based on this data a single type of forecast can be achieved. Manipulating all of these individual data and then formulating one combined forecast is the real problem.

B. Machine Learning

Before diving into deep learning which is a methodology that combines all the data and gives the desired result, it is very vital to understand where does it come from. At this point a brief introduction of Machine Learning is very necessary.

Machine learning is a collection of computer programs whose main goal is to create mathematical models from samples using statistics. Learning the execution of a computer program that utilizes training data or experiences to maximize the parameters of a model given a model that defines specific parameters. The model can forecast the future state, or it can explain the knowledge derived from the data, or it can do both. The complete process of Machine learning can be seen in Figure 2. It can be further described as follows

- defining a problem to an unknown mapping f and set a hypothetical set H of the solving model.
- Collect and organize a training set D with a finite set.
- Specify the loss function for the model.
- Select the learning algorithm.
- Obtain the parameters that make the loss function fetch the pole hour and choose them as the optimal parameters of the model.
- Save this model g with the optimal parameters, and use it to make predictions and analysis of new data.

[1]

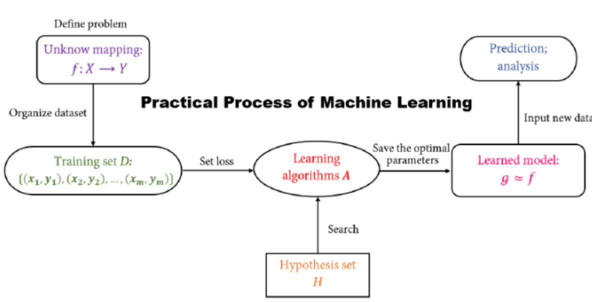

Fig. 2. This is the practical process of machine learning. [1]

According to the learning tasks, machine learning algorithms may be split into different categories, such as prediction, feature selection, and dimensional reduction. As this paper mainly focuses on forecasting modeling so only predictive aspect of Machine Learning is highlighted here.

C. Recurrent Neural Network

A neural network(NN) in easy terms is defined as a computer system that has been modelled on the human brain and nervous System. There are multiple types of Neural Networks which are used according to the requirement of prediction.

Recurrent Neural Network (RNN) is a special type of NN which is known as being an artificial type and is commonly used for time series predictions. [23] The phrase "recurrent neural network" refers to a kind of network that has an infinite impulse response, whereas "convolutional neural network" refers to a type of network that has a limited impulse response. The behavior of both types of networks is temporally dynamic. An infinite impulse recurrent network is a directed cyclic graph that cannot be unrolled and replaced with a strictly feedforward neural network, whereas a finite impulse recurrent network is a directed acyclic graph that can be unrolled and replaced with a strictly feedforward neural network. [24]The weight acquired from the input layer is the first layer. The preceding layer's weight will be applied to each subsequent layer. This Elman network features an activation function that may be any function, both continuous and discontinuous. In the present time, the delay that occurs in the first concealed layer in the prior time (t-l) can be employed (t). The feedback link, which communicates interference information (noise) from the previous input and accommodates it to the following input, is the recurrent neural network's distinguishing feature. [19] The Model of Recurrent Neural Network is described in Figure 3.

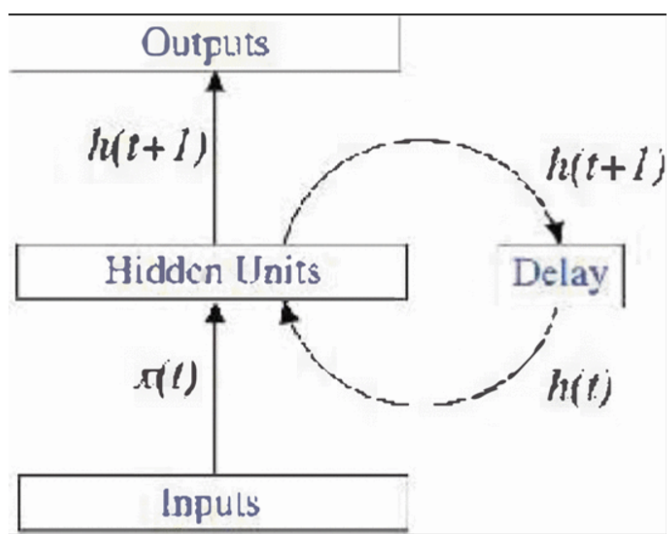

Fig. 3. Model of Recurrent Neural Networks. [19]

Let *x(t)* and *y(t)* be input and output time series. The three connection weight matrices are W_{IH}, W_{HH} and W_{HO}. The RNN parameters are learned using Back-propagation Through Time (BPTT) as described in Figure 4.

Training sequence starts at time t_0 ends at time t_1 and the sum over time of the standard error function $E_{sse/ct}$(t) at each

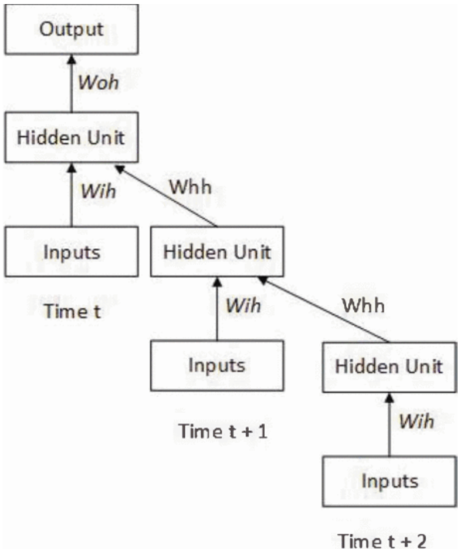

Fig. 4. Unfolded recurrent neural network model. [19]

time-step:

$$E_{total}(t_0, t_1) = \Sigma_{t=t_0}^{t_1} E_{sse/ce}(t) \tag{1}$$

$$\Delta W_{ij} = -\eta \frac{\partial E_{total}(t_0, t_1)}{\partial W_{ij}} = -\eta \Sigma_{t=t_0}^{t_1} \frac{\partial E_{total}(t_o, t_1)}{\partial W_{ij}} \tag{2}$$

[19]

D. Convolution Neural Network (CNN)

Another class of Artificial Neural Network in deep learning is Convolutional Neural Network, it is known as CNN or ConvNet. It is mostly applied to analyze visual imagery. Based on the shared-weight design of the convolution kernels or filters that slide along input features and give translation equivariant responses known as feature maps, they are also known as shift invariant or space invariant artificial neural networks (SIANN).Shift invariant or space invariant artificial neural networks are based on the shared-weight architecture of convolution kernels or filters that slide along input features and generate translation equivariant outputs known as feature maps (SIANN). [25]

A deep architecture of CNN is defined by LeCnnet al.in [26]. There might be various steps to CNN. In a CNN application using color pictures as input, the domain in feature map is normally a 2D array holding a color channel of the input image (for audio input, each feature map is a 1D array, and for video or volumetric images, it is a 3D array).

The Predictive Sparse Decomposition Algorithm is a typical unsupervised training approach for CNN to learn the filter coefficients in the filter bank layers. [27]

III. Challenges of Weather Forecasting with Deep Learning

Weather forecasting is essentially a prediction of spatiotemporal features based on a diverse array of observations from ground-based, airborne and satellite platforms.A weather forecast may be characterized as a function that maps observation data to a final forecast product if the fundamental element of the NWP workflow(Figure 1), i.e. model prediction and output post processing, is treated as a single entity.

A map of a specific weather variable (e.g. temperature), a series data of one or more variables at a specific location or aggregated over a region, some aggregate statistics of a specific variable over a given time range, a (categorical) warning index, and so on can all be used to create a forecast product. With the existing NWP, we are accustomed to using a single forecasting system (which may have many components) to generate the whole range of forecast products demanded by end users or required for system assessment and development. DL applications, on the other hand, flourish when they can focus on a single goal, i.e. a narrow collection of target variables. As a result, an end-to-end DL weather forecast, as shown in the right column of (Figure 1), would most likely be made up of numerous deep NNs, each trained on a subset of forecast items.

Data processing is a stumbling block to further scaling of NWP models and other applications on existing and future supercomputing platforms. It is currently hard to estimate how much processing time would be saved if all weather forecasting relied on DL. A good analogy should always take into account the complete weather prediction workflow as well as the entire spectrum of forecast products to be produced. Some other challenges faced by Deep Learning for Weather Forecasting are described in Figure 5.

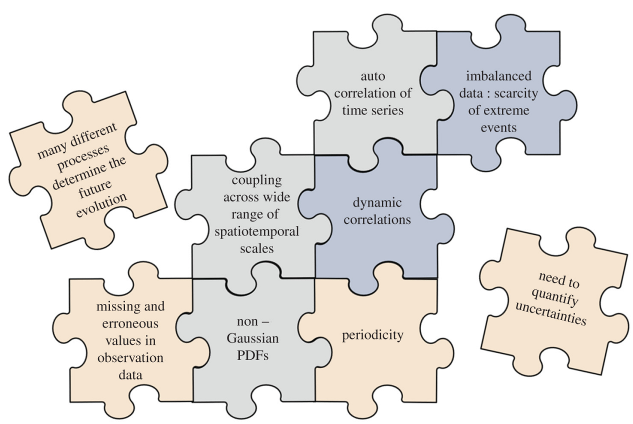

Fig. 5. Challenges faced by Deep Learning in Forecasting Weather [2]

IV. Research Methodology

A. Research Framework

The framework for this research paper can be described by the following diagram.

B. Dataset

It was not an easy task finding the dataset for Weather forecasting but after a lot of research and browsing through alot of different websites like FiveThirtyEight, Socrata, Github

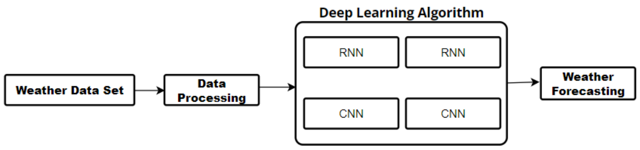

Fig. 6. Framework of Research

repositories I came across Kaggle from which I found the dataset by a person named Dheeraj Balchandani [28]. I edited the dataset as it contained a lot of irrelevant information that was a bit too much. I reduced the file and then made a new csv file with less data so that it will work easily with my PC. The data contains different observations taken at different times which describe the type of weather it is and its attributes. There are 11 coloumns in the data set which include Summary, Precip Type, Temperature, Apparent Temperature, Humidity, Wind Speed, Wind Bearing, Visibility, Loud Cover, Pressure and Daily Summary. Before being used as training dataset, the data is normalized.

	Summary	Precip Type	Temperature (C)	Apparent Temperature (C)	Humidity	Wind Speed (km/h)	Wind Bearing (degrees)	Visibility (km)	Loud Cover	Pressure (millibars)	Daily Summary
0	Partly Cloudy	rain	9.472222	7.388889	0.89	14.1197	251.0	15.8263	0.0	1015.13	Partly cloudy throughout the day.
1	Partly Cloudy	rain	9.355556	7.227778	0.86	14.2646	259.0	15.8263	0.0	1015.63	Partly cloudy throughout the day.
2	Mostly Cloudy	rain	9.377778	9.377778	0.89	3.9284	204.0	14.9569	0.0	1015.94	Partly cloudy throughout the day.
3	Partly Cloudy	rain	8.288889	5.944444	0.83	14.1036	269.0	15.8263	0.0	1016.41	Partly cloudy throughout the day.
4	Mostly Cloudy	rain	8.755556	6.977778	0.83	11.0446	259.0	15.8263	0.0	1016.51	Partly cloudy throughout the day.

Fig. 7. Dataset

C. Implementation of Weather Forecasting Data

After the data set .csv was edited and completed the Next step is to implement it in Python code. For this I am using Anaconda IDE, Kaggle Notebook and Jupyter Notebook. The libraries I am using for the implementation are **numpy**, **pandas** and **matplotlib**.

For Importing the .csv file following code was used.

```
data = pd.read_csv("C:\\Users\\acer\\Downloads\\
    archive\\weatherHistory.csv")
```

Listing 1. Importing the .csv

For Preparing the data for training and for removing all the not required fields following code was executed.

```
data.drop(['Daily Summary','Loud Cover','Wind
    Bearing (degrees)'],axis=1,inplace=True)
```

Listing 2. Preparing Data for training the model

Then for converting categorical data to numerical data

```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
data['Precip Type']=le.fit_transform(data['Precip
    Type'])
data['Summary']=le.fit_transform(data['Summary'])
```

Listing 3. Categorical to Numerical Conversion [28]

Then by declaring dependent and independent variable their correlation was determined. By this we found out which

	Precip Type	Temperature (C)	Apparent Temperature (C)	Humidity	Wind Speed (km/h)	Visibility (km)	Pressure (millibars)
Precip Type	1.000000	-0.567672	-0.570158	0.239431	-0.071101	-0.324182	0.009499
Temperature (C)	-0.567672	1.000000	0.992545	-0.633583	0.012059	0.391344	-0.005401
Apparent Temperature (C)	-0.570158	0.992545	1.000000	-0.603649	-0.054444	0.380201	-0.000021
Humidity	0.239431	-0.633583	-0.603649	1.000000	-0.226718	-0.367396	0.005065
Wind Speed (km/h)	-0.071101	0.012059	-0.054444	-0.226718	1.000000	0.107510	-0.049800
Visibility (km)	-0.324182	0.391344	0.380201	-0.367396	0.107510	1.000000	0.060925
Pressure (millibars)	0.009499	-0.005401	-0.000021	0.005065	-0.049800	0.060925	1.000000

Fig. 8. Correlation of Variables

variable can be dropped which is evident in Figure 8. From this it is evident that Temperature and apparent temperature are quite close to each other so we can drop one of them.

Afterwards, we split the dataset into train data and test data.

```
from sklearn.model_selection import train_test_split

x_train,x_test,y_train,y_test = train_test_split(x,y
    ,test_size=0.3,random_state=1)
```

Listing 4. Splitting data [28]

Then we train data by the following line of code.

```
from sklearn.ensemble import RandomForestClassifier

RF = RandomForestClassifier(max_depth=32,
    n_estimators=120,random_state=1)
RF.fit(x_train,y_train)
y_pred = RF.predict(x_test)in_test_split(x,y,
    test_size=0.3,random_state=1)
```

Listing 5. Training the data [28]

In the end we measure the accuracy of the model using the dataset.

```
from sklearn.metrics import accuracy_score
accuracy_score(y_test, y_pred)
```

Listing 6. Training the data [28]

The accuracy of the model is 56% which is still good compared to the amount of data provided.

V. Conclusion

In this paper, I highlighted the promise of current data-driven end-to-end weather forecast applications using modern DL methods. While there have been some spectacular success stories from DL applications in other domains, and preliminary attempts to apply DL to meteorological data have been performed, this study is still in its early stages.Weather data has unique qualities that need the creation of new methodologies beyond the traditional ideas of computer vision, speech recognition, and other common machine learning tasks. DL solutions for several of these concerns are being developed, but there is presently no DL approach that can deal with all of these issues at the same time, as would be necessary in a full weather forecast system.

As more advanced ML architectures become available and can be easily applied on current computer systems, I predict the area of ML in weather and climate science to develop substantially in the next years. At this time, it is impossible to say if DL will advance enough to replace most or all of the present Numerical Weather Prediction systems.

VI. Acknowledgement

I **Muhammad Moaz Amin** herewith declare that I have composed the present paper and work by myself and without use of any other than the cited sources and aids. Sentences or parts of sentences quoted literally are marked as such; other references with regard to the statement and scope are indicated by full details of the publications concerned. The paper and work in the same or similar form has not been submitted to any examination body and has not been published. This paper was not yet, even in part, used in another examination or as a course performance.

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